(i)

$$y_n = (-1)^n \frac{1}{z}\frac{d^n z}{dx^n}, \qquad y_{n+1} = -(-1)^n \frac{1}{z}\frac{d^{n+1} z}{dx^{n+1}}$$

$$\begin{aligned}
\frac{dy_n}{dx} &= (-1)^n \left(\frac{1}{z}\frac{d^{n+1}z}{dx^{n+1}} + \frac{d^n z}{dx^n} \cdot \frac{d}{dx}\left(\frac{1}{z}\right)\right) \\
\frac{dy_n}{dx} &= (-1)^n \left(\frac{1}{z}\frac{d^{n+1}z}{dx^{n+1}} + \frac{2x}{z}\frac{d^n z}{dx^n}\right) \\
&= 2x \cdot (-1)^n \frac{1}{z}\frac{d^n z}{dx^n} + (-1)^n \frac{1}{z}\frac{d^{n+1}z}{dx^{n+1}} \\
&= 2x\, y_n - y_{n+1}
\end{aligned}$$

(ii) Base case $(n = 1)$, $\qquad y_0 = 1, \quad y_1 = 2x, \quad y_2 = 4x^2 - 2$

$$\text{LHS} = y_2 = 4x^2 - 2$$

$$\text{RHS} = 2x\, y_1 - 2y_0 = 4x^2 - 2$$

$$\text{LHS} = \text{RHS}$$

Assume $y_{n+1} = 2xy_n - 2ny_{n-1}$ is true for $n = k$
Differentiate both sides:

$$\frac{dy_{k+1}}{dx} = 2y_k + 2x\frac{dy_k}{dx} - 2k\frac{dy_{k-1}}{dx}$$

Using part (i)

$$\begin{aligned}
2xy_{k+1} - y_{k+2} &= 2y_k + 2x(2xy_k - y_{k+1}) - 2k(2xy_{k-1} - y_k) \\
y_{k+2} &= 4xy_{k+1} - 2y_k - 4x^2 y_k + 4kxy_{k-1} - 2ky_k
\end{aligned}$$

$\because y_{k+1} = 2xy_k - 2ky_{k-1} \quad \implies 2xy_{k+1} = 4x^2 y_k - 4kxy_{k-1}$

$$\begin{aligned}
\therefore y_{k+2} &= 2xy_{k+1} + 4x^2 y_k - 4kxy_{k-1} - 2y_k - 4x^2 y_k + 4kxy_{k-1} - 2ky_k \\
&= 2xy_{k+1} - 2(k+1)y_k
\end{aligned}$$

From mathematical induction, the equation holds true for all $k \geq 1$

Using: $y_{n+1} = 2xy_n - 2ny_{n-1}$ and $y_{n+2} = 2xy_{n+1} - 2(n+1)y_n$

$$\begin{aligned}
(y_{n+1})^2 &= 2xy_n \cdot y_{n+1} - 2ny_{n-1}y_{n+1} \\
y_n y_{n+2} &= 2xy_{n+1} \cdot y_n - 2(n+1)y_n^2
\end{aligned}$$

$$\implies y_{n+1}^2 - y_n y_{n+2} = 2(n+1)y_n^2 - 2ny_{n-1}y_{n+1} = 2n(y_n^2 - y_{n-1}y_{n+1}) + 2y_n^2$$

(iii) Let $A_n = y_n^2 - y_{n-1}y_{n+1}$

Base case: $A_1 = y_1^2 - y_0 y_2 = (2x)^2 - (4x^2 - 2) = 2 > 0$
Assume $A_k > 0$ for some $k \geq 1$
Using (ii)

$$A_{k+1} = 2kA_k + 2y_k^2$$

Since $A_k > 0$ and $k \geq 1$, $2kA_k > 0$ and $2y_k^2 \geq 0$ Hence $A_{k+1} > 0$
From mathematical induction, $A_n > 0$ for $n \geq 1$

This solution was submitted by **Jon**

Published and refined by **Jon**

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